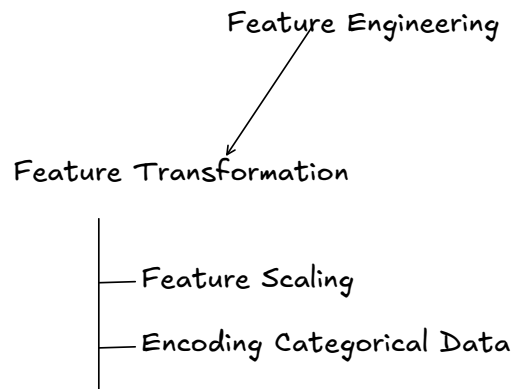
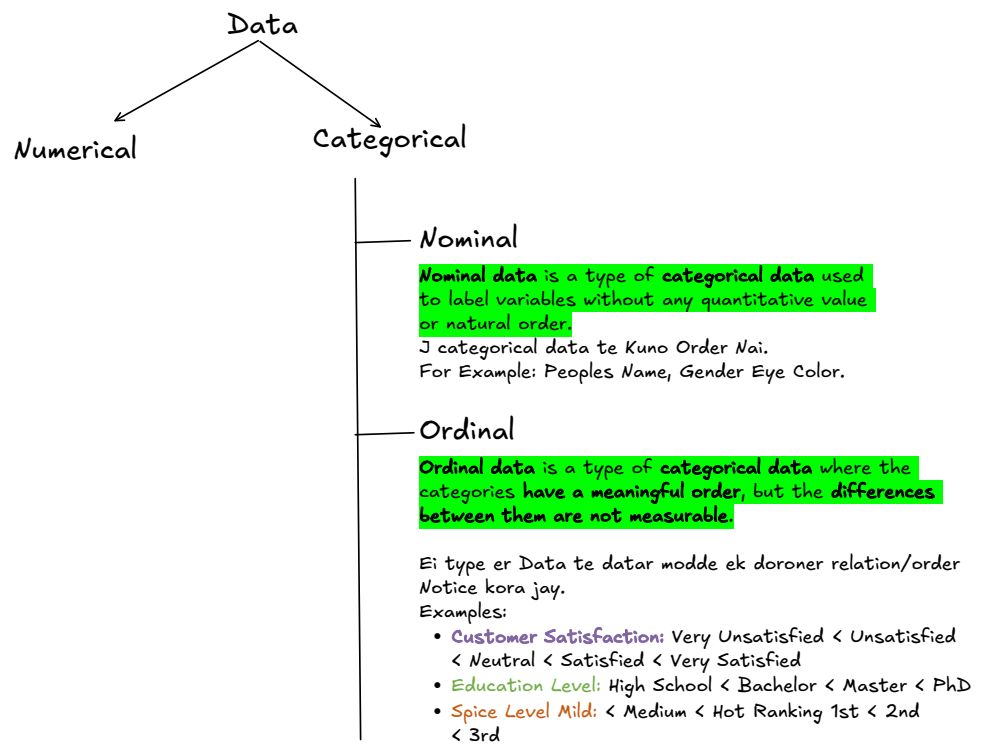


24_Encoding Categorical Data

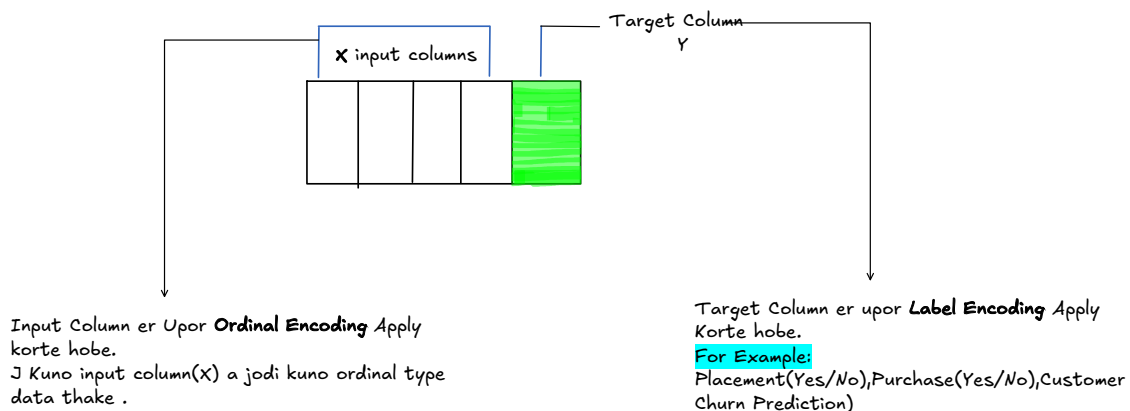
Saturday, July 5, 2025 10:59 AM



Types of Data:



Encoding Categorical Data:



Ordinal Encoding: is a method used to convert ordinal categorical data (which has a meaningful order) into numerical values that reflect that order.

→ **Why Use Ordinal Encoding?**

Machine learning models work with *numbers* — they can't understand text labels like "Low", "Medium", "High". So, ordinal encoding converts these labels into integers that preserve the order.

→ **Use in Machine Learning:**

Ordinal encoding is also useful when your machine learning model can understand order in numbers. For example:

- **Linear Regression:** It looks at the relationship between input and output values — so ordered numbers help.
- **Decision Trees or Gradient Boosting:** These models split data based on values, so having ordered numbers helps them make smarter decisions.

Label Encoding :

Label encoding is most commonly used for the target variable (output column) in classification problems.